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name: <unnamed>
log: D:\Financial Modeling\Bahria Fall 2025\Panel Data\Panel Data Revision Class Tests.smc
log type: smcl
opened on: 25 Dec 2025, 15:14:43

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1 . encode company, gen (comp)

2 . tsset comp year
   panel variable:  comp (unbalanced)
   time variable:  year, 2007 to 2010, but with gaps
               delta: 1 unit

3 . reg cash cft liq lev

```

Source	SS	df	MS	Number of obs = 1005		
Model	3.65147925	3	1.21715975	F(3, 1001) = 194.17		
Residual	6.27470853	1001	.00626844	Prob > F = 0.0000		
				R-squared = 0.3679		
				Adj R-squared = 0.3660		
Total	9.92618778	1004	.009886641	Root MSE = .07917		

cash	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
cft	-.0046202	.0007308	-6.32	0.000	-.0060543	-.0031861
liq	-.0004881	.0000413	-11.81	0.000	-.0005693	-.000407
lev	-.0004939	.0000656	-7.53	0.000	-.0006226	-.0003651
_cons	.041385	.0025032	16.53	0.000	.0364729	.046297

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4 . hettest
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Breusch-Pagan / Cook-Weisberg test for heteroskedasticity
Ho: Constant variance
Variables: fitted values of cash

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chi2(1)      = 2.31
Prob > chi2   = 0.1282

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5 . winsor cash, gen(wcash) p(0.1)

6 . winsor cft , gen(wcf) p(0.1)

7 . winsor liq , gen(wliq) p(0.1)

8 . winsor lev , gen(wlev) p(0.1)

9 . reg wcash wcf wliq wlev

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Source	SS	df	MS	Number of obs = 1005		
Model	.364855127	3	.121618376	F(3, 1001) = 129.14		
Residual	.942680621	1001	.000941739	Prob > F = 0.0000		
				R-squared = 0.2790		
				Adj R-squared = 0.2769		
Total	1.30753575	1004	.001302326	Root MSE = .03069		

wcash	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
wcf	.0919833	.0126867	7.25	0.000	.0670876	.1168789
wliq	.0435112	.0049957	8.71	0.000	.033708	.0533145
wlev	-.0445275	.0051903	-8.58	0.000	-.0547126	-.0343424
_cons	.0289202	.0024758	11.68	0.000	.0240617	.0337786

10 . xtreg wcash wcf wliq wlev, fe

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Fixed-effects (within) regression               Number of obs   =    1005
Group variable: comp                          Number of groups =    313

R-sq:  within = 0.0374                        Obs per group:  min =     1
        between = 0.3130                        avg   =    3.2
        overall  = 0.2392                        max   =     4

corr(u_i, Xb)  = 0.3820                        F(3, 689)       =    8.93
                                                Prob > F        =    0.0000

```

wcash	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
wcf	.0230876	.0135026	1.71	0.088	-.0034235	.0495987
wliq	.0039319	.0066769	0.59	0.556	-.0091776	.0170415
wlev	-.0259296	.0053912	-4.81	0.000	-.0365148	-.0153444
_cons	.0329275	.0023173	14.21	0.000	.0283776	.0374774
sigma_u	.02886924					
sigma_e	.01981656					
rho	.67972712	(fraction of variance due to u_i)				

F test that all u_i=0: F(312, 689) = 5.49 Prob > F = 0.0000

11 . estimates store fe

12 . xtreg wcash wcf wliq wlev, re

```

Random-effects GLS regression               Number of obs   =    1005
Group variable: comp                          Number of groups =    313

R-sq:  within = 0.0320                        Obs per group:  min =     1
        between = 0.3559                        avg   =    3.2
        overall  = 0.2758                        max   =     4

corr(u_i, X)   = 0 (assumed)                  Wald chi2(3)    =   151.58
                                                Prob > chi2     =    0.0000

```

wcash	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
wcf	.0518251	.0119652	4.33	0.000	.0283737	.0752765
wliq	.0280982	.0051941	5.41	0.000	.017918	.0382784
wlev	-.0377539	.0047116	-8.01	0.000	-.0469886	-.0285193
_cons	.0316261	.0025003	12.65	0.000	.0267256	.0365266
sigma_u	.02248722					
sigma_e	.01981656					
rho	.56288006	(fraction of variance due to u_i)				

13 . estimates store re

14 . hausman fe re

	Coefficients		(b-B) Difference	sqrt(diag(V_b-V_B)) S.E.
	(b) fe	(B) re		
wcf	.0230876	.0518251	-.0287375	.0062572
wliq	.0039319	.0280982	-.0241663	.0041956
wlev	-.0259296	-.0377539	.0118244	.0026203

b = consistent under Ho and Ha; obtained from xtreg
B = inconsistent under Ha, efficient under Ho; obtained from xtreg

Test: Ho: difference in coefficients not systematic

chi2(3) = (b-B)'[(V_b-V_B)^(-1)](b-B)
= 53.54
Prob>chi2 = 0.0000

15 . Fixed EFFECT RESULTS ARE ACCEPTED

unrecognized command: Fixed
r(199);

16 . xtabond wcash wcf wliq wlev, lags(1) twostep artests(2)

Arellano-Bond dynamic panel-data estimation Number of obs = 282
Group variable: comp Number of groups = 143
Time variable: year
Obs per group: min = 1
avg = 1.972028
max = 2

Number of instruments = 7 Wald chi2(4) = 3.27
Prob > chi2 = 0.5138

Two-step results

wcash	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
wcash						
L1.	.3862835	.262952	1.47	0.142	-.1290929	.9016599
wcf	.0066588	.0236812	0.28	0.779	-.0397554	.0530731
wliq	.0049686	.013133	0.38	0.705	-.0207715	.0307088
wlev	.0048848	.0106697	0.46	0.647	-.0160274	.0257969
_cons	.0144777	.0114396	1.27	0.206	-.0079434	.0368988

Warning: gmm two-step standard errors are biased; robust standard errors are recommended.

Instruments for differenced equation
GMM-type: L(2/.) .wcash
Standard: D.wcf D.wliq D.wlev
Instruments for level equation
Standard: _cons

17 . xtabond wcash wcf wliq wlev, inst(1.wcf 1.wlev) lags(1) twostep artests(2)

Arellano-Bond dynamic panel-data estimation Number of obs = 282
Group variable: comp Number of groups = 143
Time variable: year
Obs per group: min = 1
avg = 1.972028
max = 2

Number of instruments = 9 Wald chi2(4) = 15.14
 Prob > chi2 = 0.0044

Two-step results

wcash	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
wcash L1.	.5438299	.1487616	3.66	0.000	.2522625	.8353972
wcf	.008045	.0253777	0.32	0.751	-.0416943	.0577844
wliq	.004775	.0135934	0.35	0.725	-.0218676	.0314177
wlev	.0129532	.0079349	1.63	0.103	-.0025989	.0285054
_cons	.0073174	.0074307	0.98	0.325	-.0072466	.0218813

Warning: gmm two-step standard errors are biased; robust standard errors are recommended.

Instruments for differenced equation

GMM-type: L(2/.) .wcash

Standard: D.wcf D.wliq D.wlev L.wcf L.wlev

Instruments for level equation

Standard: _cons

18 . xtabond wcash wcf wliq wlev, inst(l.wcf l.wlev) lags(1) twostep artests(2)

Arellano-Bond dynamic panel-data estimation Number of obs = 282
 Group variable: comp Number of groups = 143
 Time variable: year

Obs per group: min = 1
 avg = 1.972028
 max = 2

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Warning: gmm two-step standard errors are biased; robust standard errors are recommended.

Instruments for differenced equation

GMM-type: L(2/.) .wcash

Standard: D.wcf D.wliq D.wlev L.wcf L.wlev

Instruments for level equation

Standard: _cons

19 . estat sargan

Sargan test of overidentifying restrictions

H0: overidentifying restrictions are valid

chi2(4) = 2.903352

Prob > chi2 = 0.5741

20 . estat abond

Arellano-Bond test for zero autocorrelation in first-differenced errors

Order	z	Prob > z
1	-2.8462	0.0044
2	.	.

H0: no autocorrelation

21 . log close

name: <unnamed>
log: D:\Financial Modeling\Bahria Fall 2025\Panel Data\Panel Data Revision Class Tests.smc
log type: smcl
closed on: 25 Dec 2025, 15:25:49
